QuarccPurple

A Market-Neutral Fund Built for AI Crowding

7.0%
a year after all costs

0.012
beta to the S&P 500

−2.99%
max drawdown

quarccpurple.ca

McGill-FIAM Asset Management Hackathon, October 2026

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Executive Summary

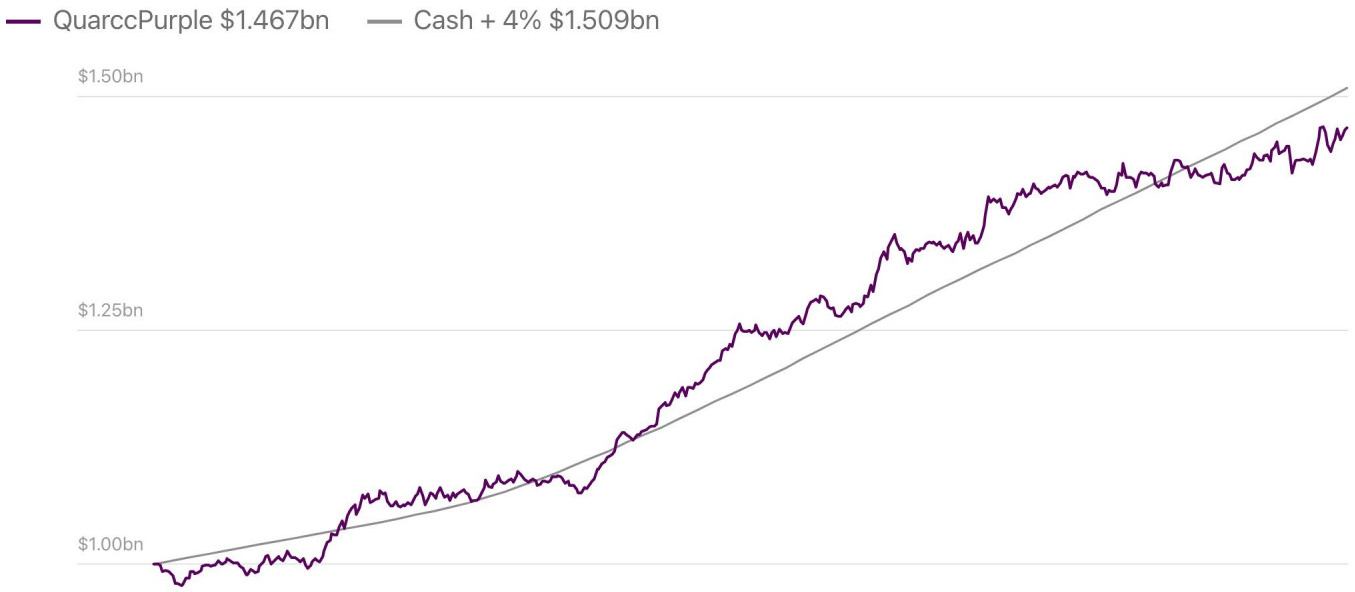
7.0%
a year after costs, cash + 4% 7.5%

0.012
beta to the S&P 500

−2.99%
max drawdown, S&P 500 −25.4%

+1.3%
in 2022, 60/40 −15.8%

Growth of $1bn, after all costs



0.00 → −0.11
information ratio before and after costs

0.57
residual Sharpe from stock picking

Exhibit 1 · Performance

4.21

Sortino ratio, S&P 500 1.51

0.52%

ulcer index, S&P 500 8.10%

71%

of months positive

METRIC	UNITS	STRATEGY	BENCHMARK
Ending value of the endowment	US$ billion	1.467	1.509
Cumulative return over the period	%	46.71	50.87
Annualised return, CAGR	% a year	7.00	7.53
Annualised volatility	% a year	3.96	0.56
Annualised alpha vs S&P 500	% a year (s.e.)	3.47 (1.42)	n/a
Beta vs S&P 500, monthly	coefficient (s.e.)	0.012 (0.030)	n/a
Beta vs S&P 500, daily	coefficient (s.e.)	0.018 (0.007)	n/a
Sharpe ratio	annualised	0.91	undefined
Information ratio vs T-bill + 4%	annualised (s.e.)	−0.11 (0.42)	n/a
Maximum drawdown	%	−2.99	0.00
Positions held	average, min, max	314, 246, 357	n/a
Gross exposure	average, max, % of capital	136.5, 192.9	n/a
Net exposure	average, min, max, % of capital	−0.3, −3.1, 2.2	n/a
One-way turnover	% a rebalance, a year	3.9, 94	n/a

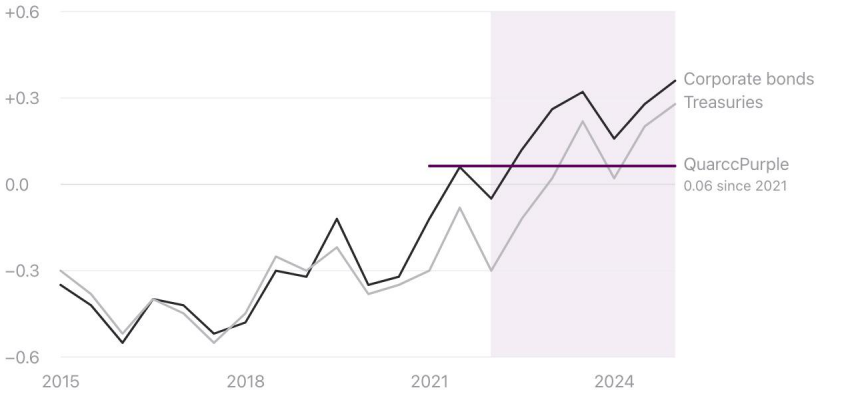
Performance During Market Shocks

The fund against the S&P 500 in its worst stretches

Fund S&P 500

	Fund	S&P 500
S&P 500's 5 worst months compounded	+0.11%	−32.40%
2022 drawdown Jan to Sep 2022	+2.24%	−23.93%
Calendar 2022 stocks and bonds fell	+1.34%	−18.18%
Average S&P 500 down month 23 of 68 months	+0.57%	−3.64%

Correlation to equities



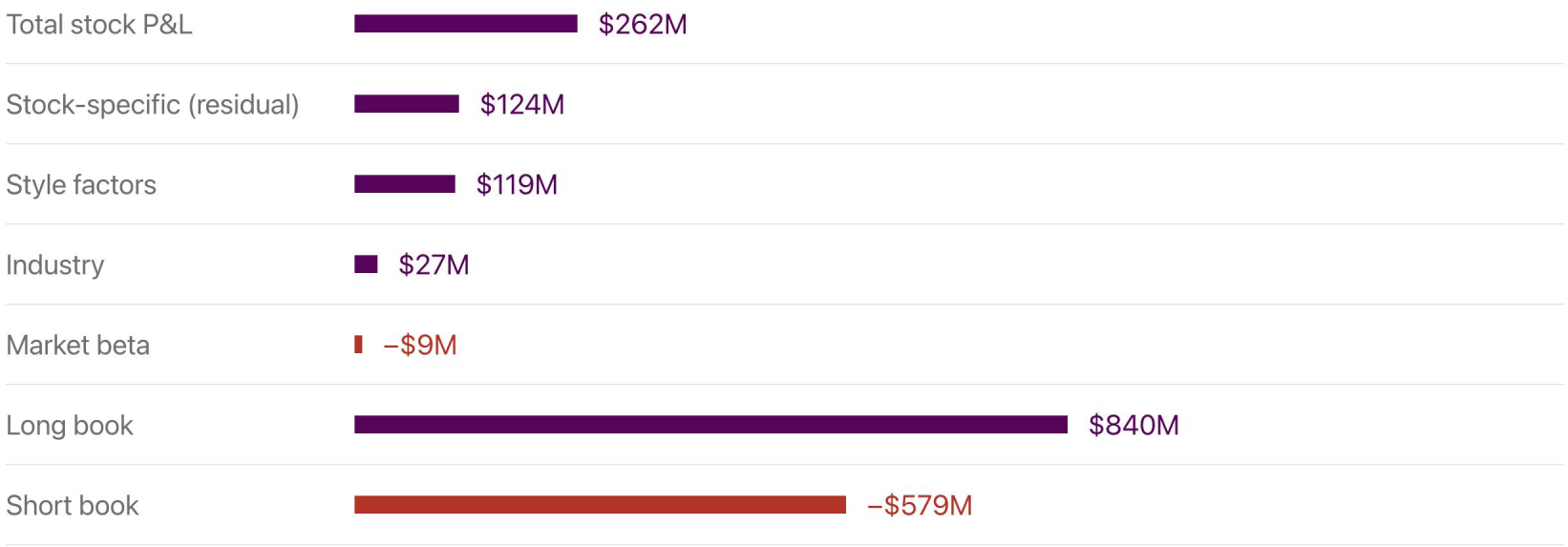
65%
of S&P 500 down months positive for us

−15%
downside capture

0.02
correlation to a 60/40

Exhibit 2 · Independent Verification by Arcana

Where the stock P&L came from, 2021 to 2026



0.57

Residual Sharpe, factors 0.83

–0.02

Average predicted beta

313

Average positions, ours 314

Machine Learning Signals

The organisers' 147

Quality, value and momentum, weighted by E-LASSO

SYY-10 mispricing

Over-priced stocks that are hard to short

Earnings surprise

Prices drift after earnings, LightGBM forecasts it

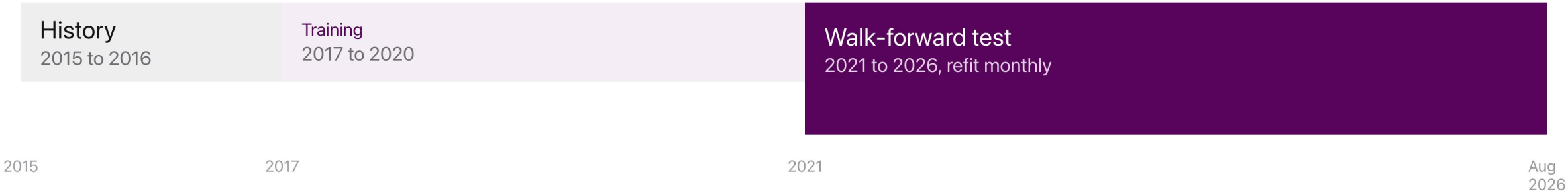
Short interest

Heavily shorted stocks lag

Mid-month refresh

Fast price signals re-read between month ends

Preparing the Data



Organisers' panel

Readable at its month-end close

Organisers' daily file

Trades at the next close

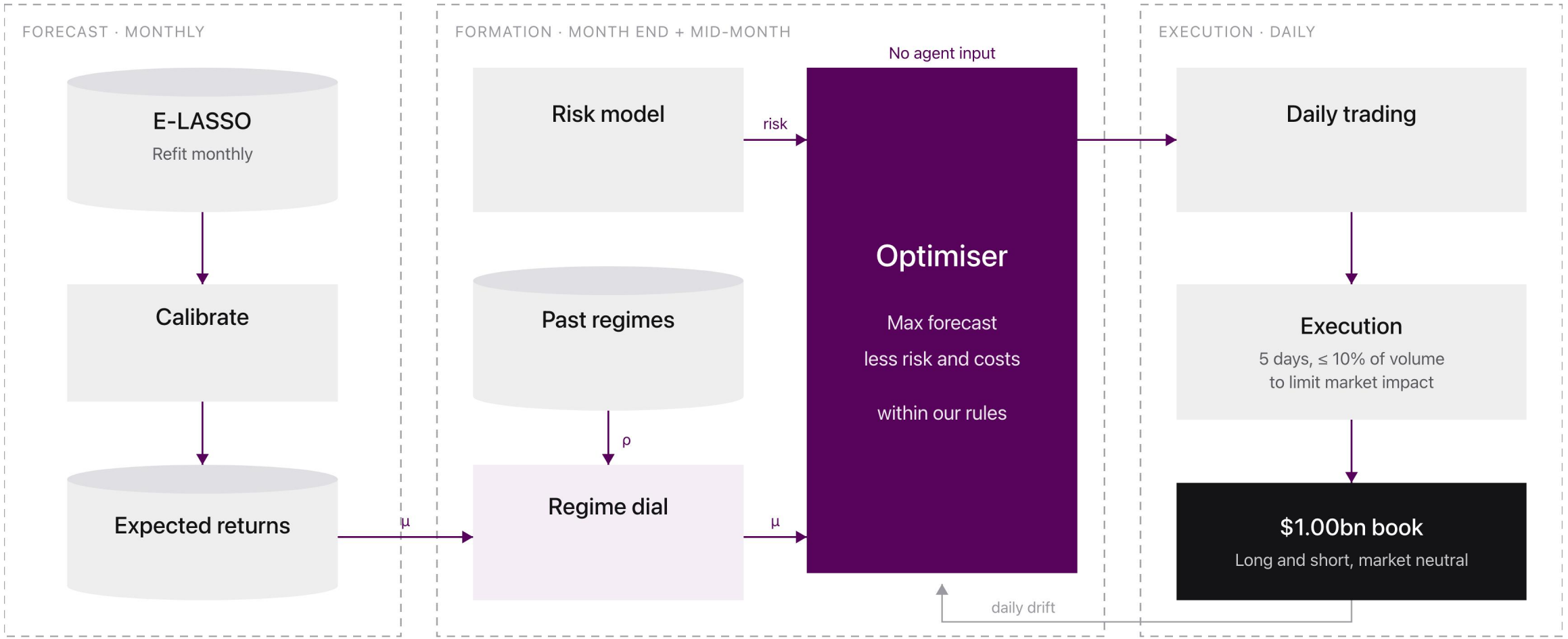
Our three signals

Trades one close later

Carbon Arc, four questions

Fails the four questions, never used

From Forecast to Book



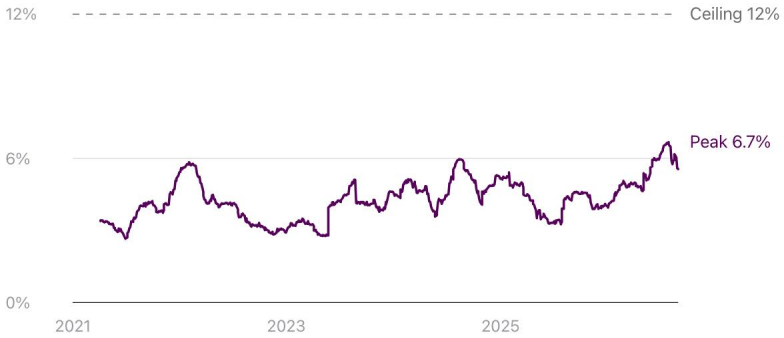
Portfolio Construction

Our rules and the hackathon limits

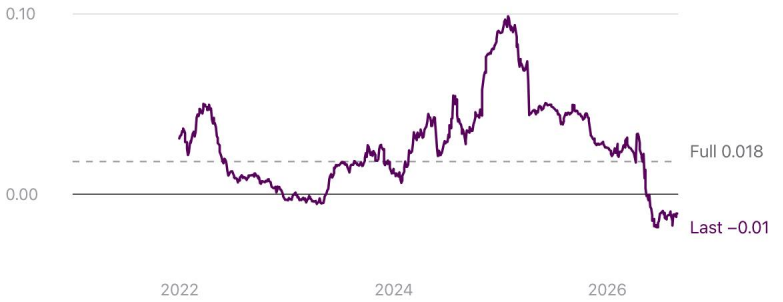
RULE	OURS	LIMIT	REALISED
Dollar net	0	±50%	−3.1% to +2.2%
Market beta	±0.02	n/a	0.01
Sector net	±1%	±20%	1.8%
Sectors held	all 11	at least 5	11
One name	3.9% long, 1.08% short	5%	4.6%
Names	200 to 500	200 to 500	246 to 357
Gross	1.8 target	200%	193%
Volatility	12% guard	12 to 14%	3.96%
Short floor	$500m	$500m	0 breaches
Rebalancing	twice a month	daily to monthly	136

0 breach days

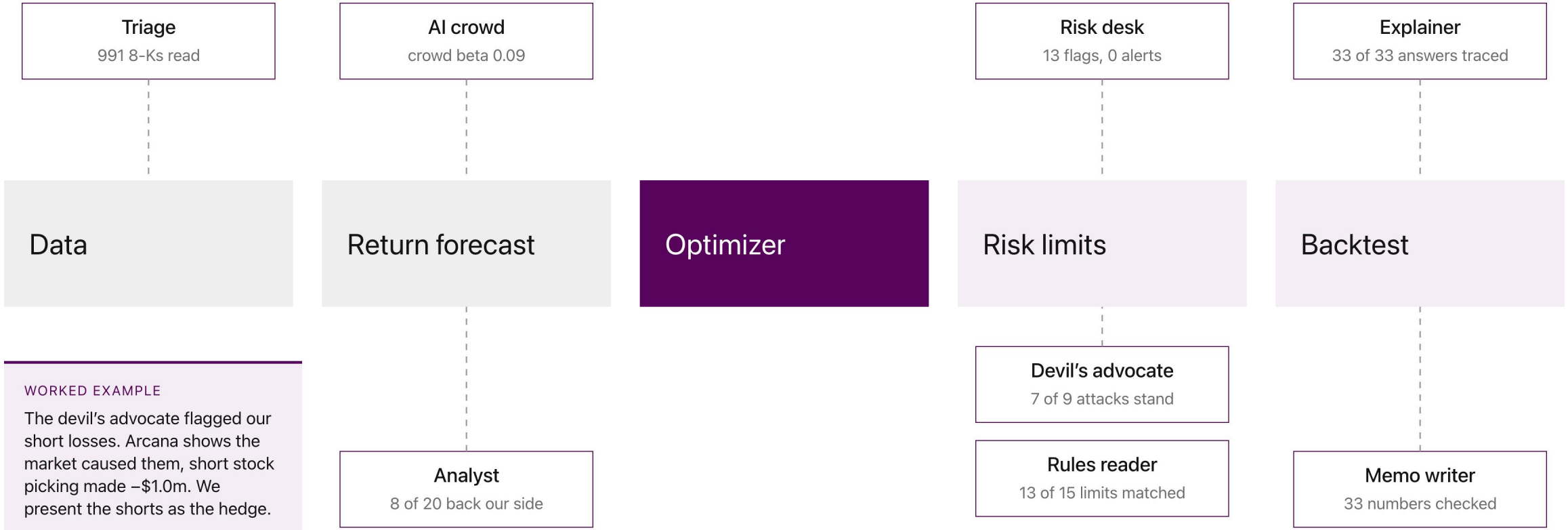
Rolling volatility, ceiling 12%



Rolling beta to the S&P 500



Our AI Agents



WORKED EXAMPLE

The devil's advocate flagged our short losses. Arcana shows the market caused them, short stock picking made –$1.0m. We present the shorts as the hedge.

What Worked and What Fell Short

2022

+1.3%

trailed the hurdle by 4.9 points

2023

+15.1%

beat the hurdle by 5.6 points

2024

+11.2%

beat the hurdle by 1.8 points

2025

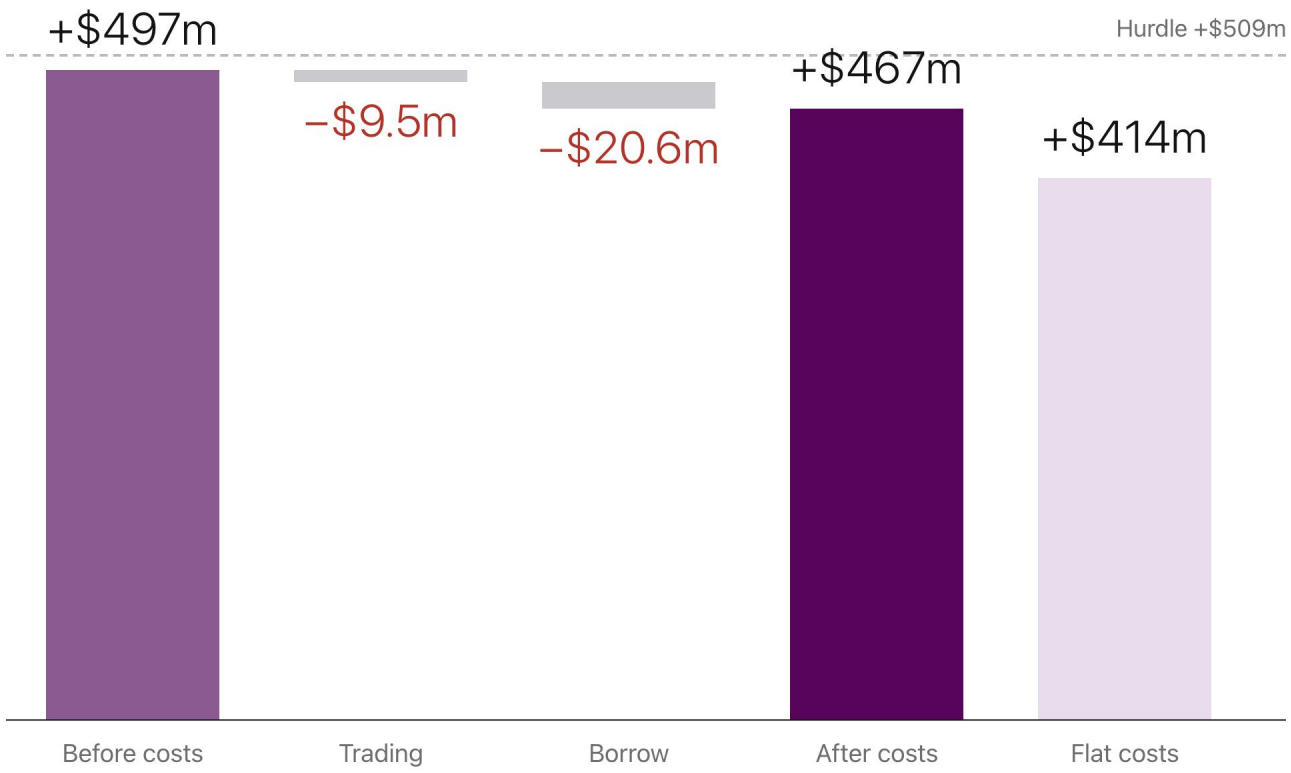
+2.1%

trailed the hurdle by 6.3 points

What drove it	2023 and 2024. Arcana splits stock P&L evenly between style and stock picking.
AI audits	An AI audit caught a filter that could read split-adjusted prices. Code blocks it.
Agents	Eight AI agents checked the book and 7 of 9 attacks stood.

Discussion and What We Learned

Gain on $1bn before and after costs, US$ millions



Discussion

1. It behaved as built. Beta 0.012, volatility 3.96%, 0 breach days. Return fell short, 7.0% a year against 7.5%.
2. Costs were small, $30m in all and mostly borrow. Before costs the fund came within $12m of the hurdle.

What we learned

1. Stock-specific alpha is modest, residual Sharpe 0.57. Much of the return is style exposure.
2. Next, a stronger short side and lower borrow cost.

Our Team



LEAD

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Concordia · Computer Science

AI & Research



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Concordia · Mathematical & Computational Finance

Risk Management



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Concordia · Finance

Portfolio Construction



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McGill · Honours Physics & Computer Science

Machine Learning



Nicolas Romero

Concordia · Finance, Mathematics & Statistics

Data & Finance

7.0% a year after all costs · beta 0.012 · max drawdown –2.99% · verified by Arcana

Risk-Adjusted Scorecard

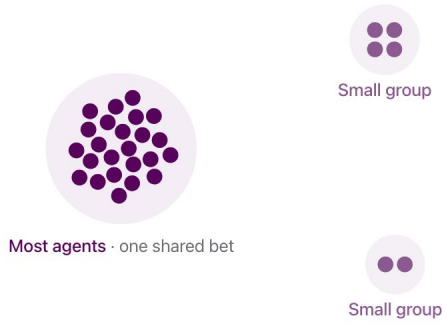
0.91 Sharpe ratio	4.21 Sortino ratio, S&P 500 1.51	2.34 Calmar ratio	+15% Upside capture
0.52% Ulcer index, S&P 500 8.10%	3.96% Volatility	−2.99% Max drawdown	−1.43% Worst month
71% Months positive	54% Days positive	1.30 Profit factor	0.06 Correlation to the S&P 500

Crises and the AI Crowd

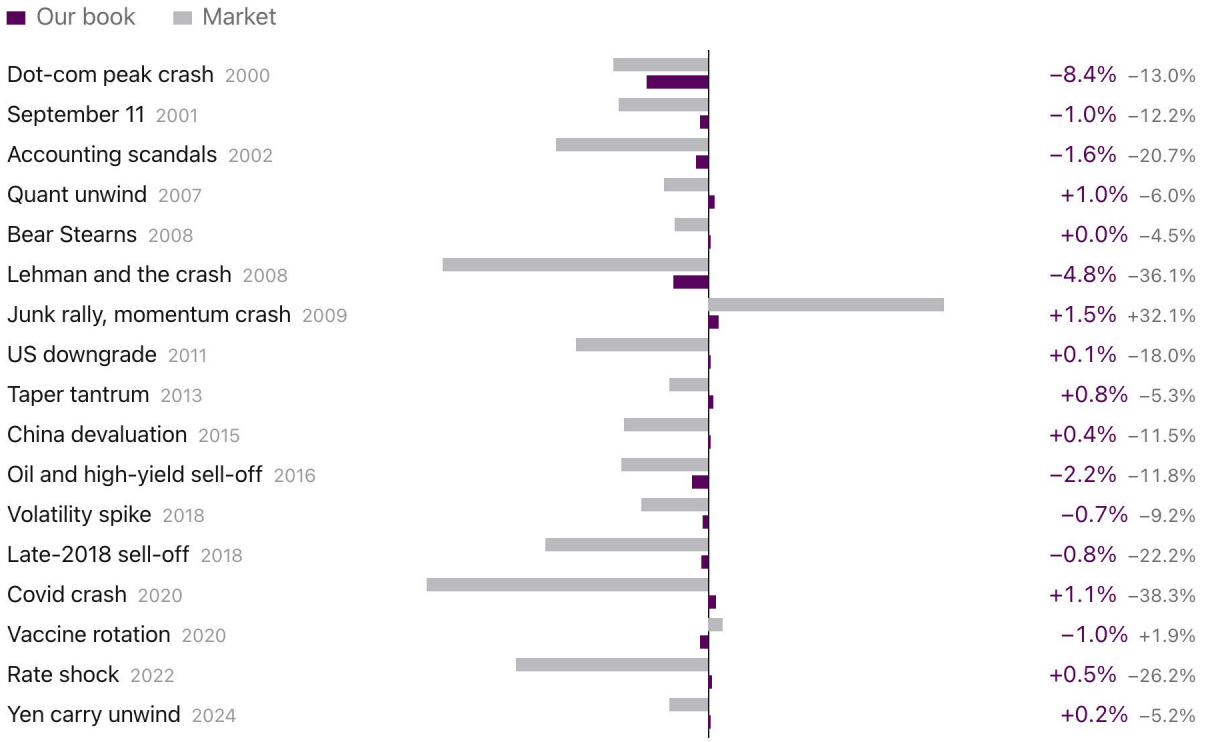
Correlation, monthly

AGAINST	CORRELATION
S&P 500	0.06
60/40 portfolio	0.02
AI crowd, long-short	0.32
AI crowd, long only	−0.12

AI crowd, 30 agents vote like 4

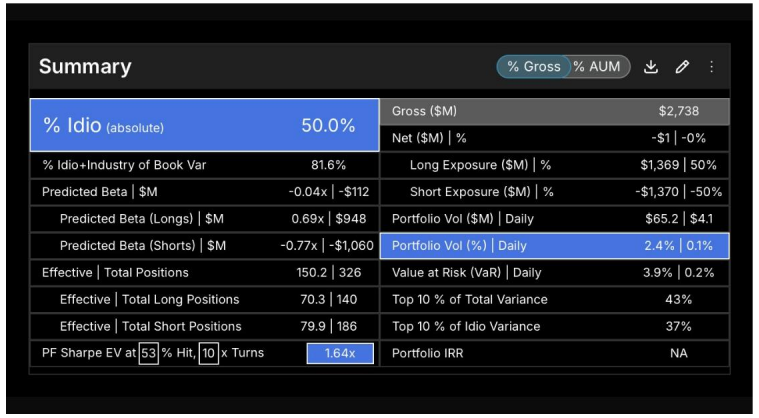


Today's book replayed through each crisis

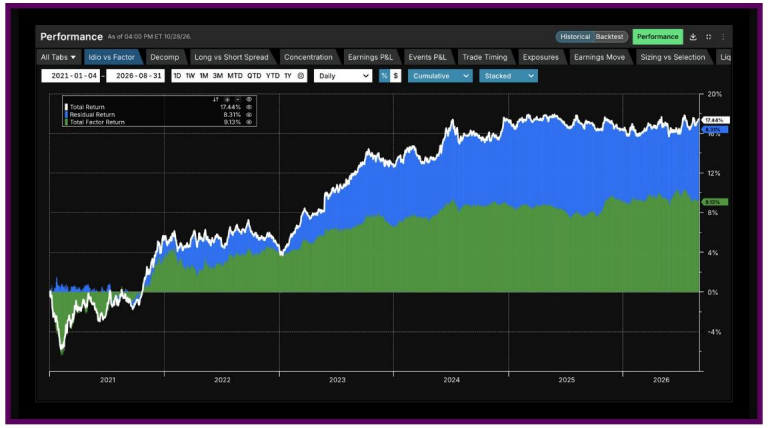


Average **−0.9%** against −12.1% for the market

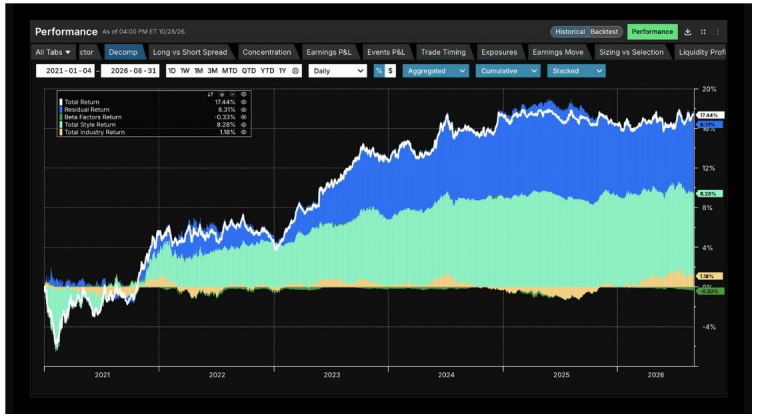
Exhibit 2 · Arcana Returns



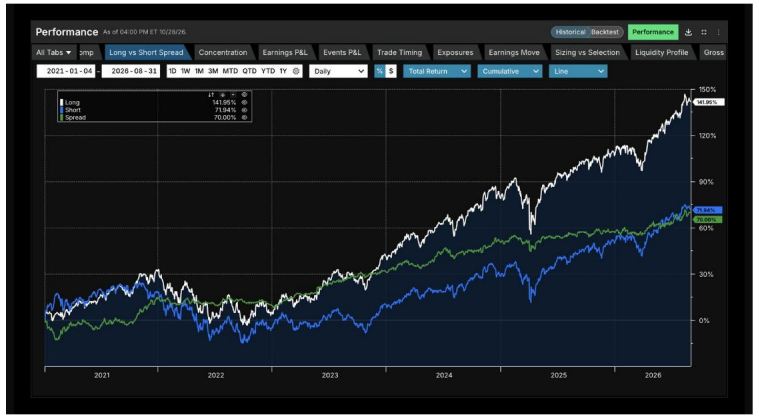
Summary · the risk snapshot, 50% stock-specific



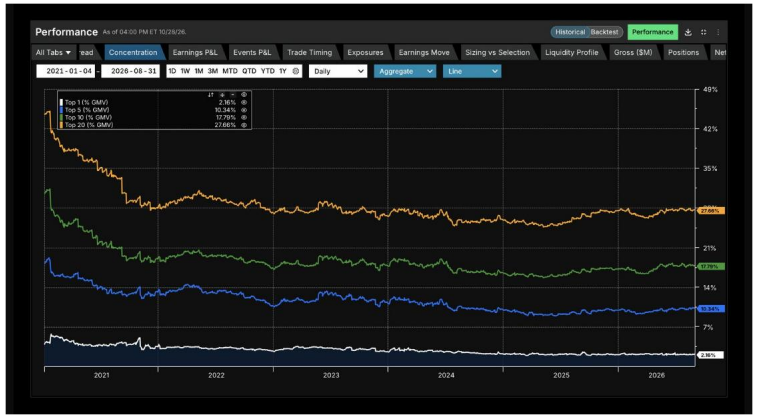
1 Idiosyncratic against factor · stock picking 8.3%, factors 9.1%



2 Decomposition · where the return came from



3 Long against short · what each leg made

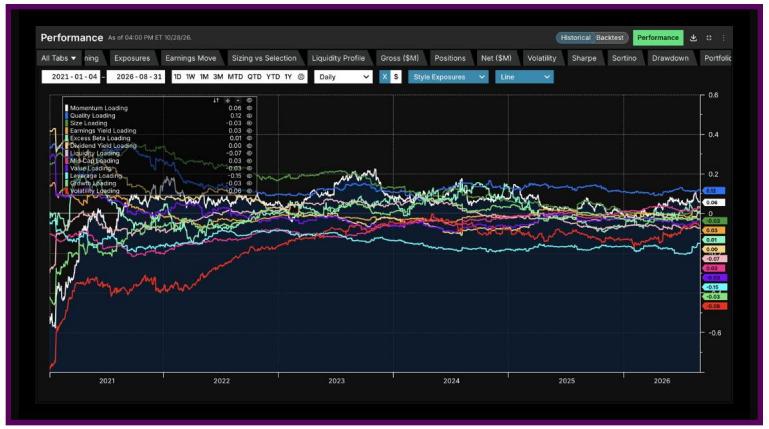


4 Concentration · share of gross in the top names

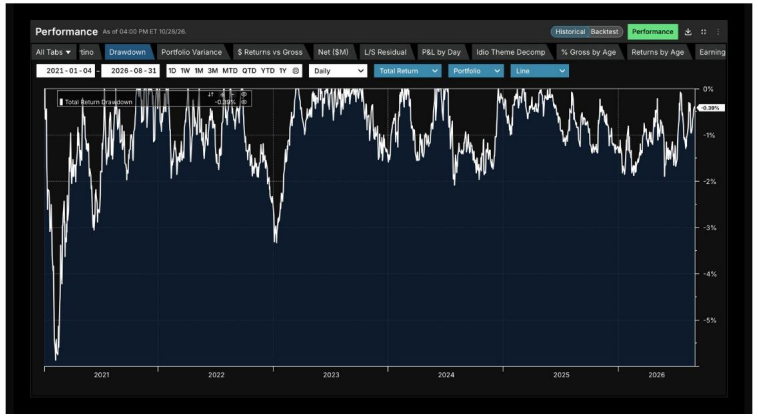


5 Earnings · inside and outside earnings windows

Exhibit 2 · Arcana Risk and Style



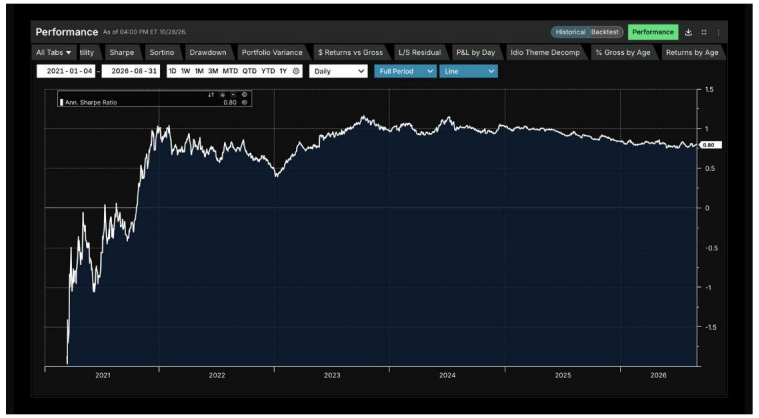
6 Style exposure · every style within ±0.15



7 Drawdown · our fund



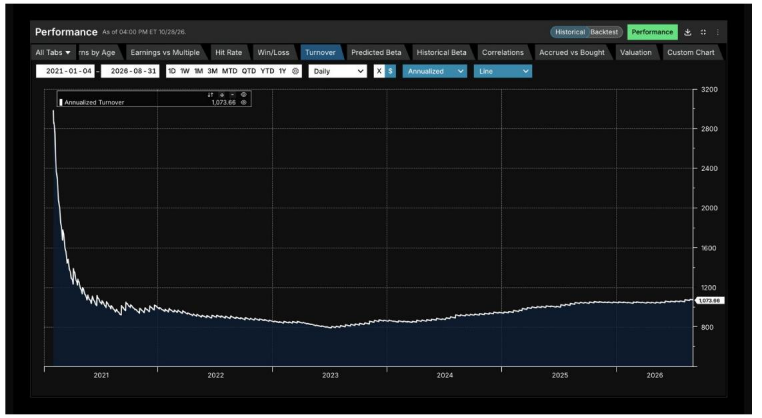
7 Drawdown · the S&P 500 benchmark



8 Sharpe ratio · risk-adjusted return



9 Hit rate · share of positions that made money



10 Turnover · trading through the book